

Aureus

A four-band EQ, drive and compressor tuned by the golden ratio.



Overview

Aureus is a multiband processor. It splits the signal into four bands - SUB, LOW-MID, HIGH-MID and PRESENCE or AIR - and gives each one an EQ, a drive stage and a compressor, then a band gain with SOLO and MUTE. The bands add back together flat and in phase, and MIX blends in a dry signal that is phase-aligned with them.

The crossovers are decadic: the second split is ten times the first, and the third is a hundred times the first, held between 5 and 10 kHz so the top band is always presence or air. The first split is picked from the Fibonacci numbers 34, 55, 89 and 144 Hz, or the classic 70 Hz.

Each section has just one knob and a type key. Everything the knobs don't set - EQ frequencies and shapes, attack and release times, ratios and knees - is worked out from the golden ratio (1.618), so the four bands relate to each other in a consistent, musical way. The compressor attack gets faster from band to band (30, 18.5, 11.5 and 7.1 ms, with the release about eleven times longer), before each compressor type adjusts the timing.

Reach for it for

- Mix-bus and mastering glue across four bands
- Adding tape, transistor or transformer colour to just the lows or the top
- Lifting air or presence with a single knob
- Taming harsh or boomy bands only when they get loud (DYNAMIC EQ)
- Parallel multiband processing with MIX

Controls

Left: display, presets and setup

Display	Shows the current preset, or the control you are touching and its value. Click it for the preset list.
PRESETS	Opens the list of factory presets; the ◀ / ▶ keys beside it step through them.
SIZE: S / M / L	Panel size: small, medium or large. The panel also shrinks to fit a smaller plugin window.
?	Opens a pop-up with the design credit, the GFX website address and the address of this manual. Click anywhere to close it.
INPUT	Input level before anything else, -24 to +24 dB. It also sets how hard the signal hits the drive and compressor stages.
CROSSOVER F1 x10 x100	Picks the first crossover: 34, 55, 70 (the default), 89 or 144 Hz. The second sits at ten times that, the third at a hundred times, held between 5 and 10 kHz (5, 5.5, 7, 8.9 and 10 kHz). The display shows all three when you choose. The top band is called PRESENCE with the two lowest settings and AIR with the others.
HW	Switches the hardware finish (shading, bevels and coloured knob caps) on or off.
GOLD / TEAL	Switches the faceplate between yellow and petrol teal.

The four bands

Each band header shows its frequency range and an activity light: green when the band is passing signal, red when it gets near clipping. Each row has a type key - click it to step through the three types - and an amount knob.

EQ type: PROGRAM	Broad, gentle curves. SUB is a low shelf below the first crossover and the top band a high shelf above the third; the two middle bands are broad bells at their centres. Boosting a shelf also adds a small dip right at the band edge, in the
-------------------------	--

	spirit of the classic passive program equalisers, so the boost stays clean.
EQ type: CONSOLE	Firmer, console-style shapes: resonant shelves, and bells that get narrower as you boost or cut further.
EQ type: DYNAMIC	The CONSOLE shape, but its gain follows the band's level: nothing below about -30 dB, rising to the full knob setting as the band gets louder. Use a cut to tame a band only when it gets loud.
EQ knob	The EQ gain, -12 to +12 dB.
DRIVE type: TRANSISTOR	A firm, mostly odd-harmonic clip with a harder edge.
DRIVE type: TAPE	Soft, slightly lopsided saturation with a touch of hysteresis; the top end rolls off a little more as the drive goes up. The default.
DRIVE type: TRANSFORMER	The lows of the band saturate hardest, like an iron-core transformer, with a cleaner top.
DRIVE knob	How hard the band is driven, 0 to 100%. The output is partly level-compensated. 0 is clean.
COMP type: OPTO	Smooth and slower, measuring the average level; the release slows as the gain reduction deepens. Up to about 2.6:1 with a wide, soft knee. The default.
COMP type: VCA	Fast and firmer, with a feedback detector and a tighter knee; up to about 4.2:1.
COMP type: GENTLE	Three gentle compressors in series at thresholds 6 dB apart, each carrying a third of the ratio (up to about 4.2:1 in total), with a release that adapts to the music. In the spirit of the COMP mode of RND.
COMP knob	How much compression, 0 to 100%. One knob does it all: turning it up lowers the threshold (from -6 down to -36 dB) and raises the ratio together, and adds some make-up gain automatically. 0 is off.

GR	The band's gain reduction, 0 to 20 dB, top to bottom.
GAIN	The band's output level, -12 to +12 dB.
SOLO	Solo in place; several bands can be soloed at once.
MUTE	Silences the band.

OUTPUT

VU meters	Left and right output level, with 0 VU at -18 dBFS.
------------------	---

MASTER

DRIFT	Golden-ratio breathing, 0 to 100% (25% by default): each band's compressor threshold wanders slowly up and down, by up to 3 dB at 100%, each on its own period (3 seconds times 1.618 to the power of the band), so the bands never move together. Only heard where the compressor is working.
MIX	Dry / processed balance, 0 to 100% (100% by default). The dry signal is phase-aligned with the bands, so partial settings blend cleanly.
OUTPUT	Output level, -24 to +12 dB.
LIMIT	A safety limiter holds the output peaks at about -0.3 dBFS. The light shows when it is working.

Drag the knobs up and down (hold Shift for fine moves); double-click or Ctrl-click to reset one.

Presets

Init	Everything flat and clean: no EQ, drive or compression.
Golden Glue	Light tape drive and GENTLE compression at 30% on every band.

Tape Polish	Tape drive on all bands with gentle opto compression and +1.5 dB on the SUB and top shelves.
Opto Smooth	Opto compression at 40% on every band, no EQ or drive.
Air Lift	+4 dB on the top shelf with a little drive and GENTLE compression, and +1 dB on HIGH-MID (CONSOLE).
Iron Low End	Transformer drive on SUB (45%) with +2.5 dB and opto compression; LOW-MID trimmed 1.5 dB (CONSOLE) with transformer drive.
VCA Punch	VCA compression on the lower three bands, with transistor drive on LOW-MID.
Dynamic Tame	DYNAMIC EQ cuts on LOW-MID, HIGH-MID and the top band, with light opto compression.

Tips

- The presets set the bands, DRIFT, MIX, INPUT and OUTPUT, and clear SOLO and MUTE; the crossover setting is kept.
- Use INPUT to set how hard the drive and compressors work, then OUTPUT to bring the level back.
- Try the crossover choices on a full mix: 55 or 70 Hz keeps kick and bass together in LOW-MID, 89 or 144 Hz puts more of the bass weight in SUB.
- For a harsh top end, set the top band to DYNAMIC and cut a few dB: it only works on the loud, bright moments.

File: gfx-aureus.jsfx